Paleomagnetism and U–Pb geochronology of the Clarence Head dykes, Arctic Canada: orthogonal emplacement of mafic dykes in a large igneous province

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Abstract: The north–south-trending Clarence Head dyke swarm, located on Devon and Ellesmere Islands in the Canadian High Arctic, has a trend orthogonal to that of the Neoproterozoic Franklin swarm that surrounds it. The Clarence Head dykes are dated by the U–Pb method on baddeleyite to between 716 ± 1 and 713 ± 1 Ma, ages apparently younger than, but within the published age range of, the Franklin dykes. Alpha recoil in baddeleyite is considered as a possible explanation for the difference in ages, but a comparison of the U–Pb ages of grains of equal size from both swarms suggests that recoil distances in baddeleyite are lower than those in zircon and that the Clarence Head dykes are indeed a distinctly younger event within the period of Franklin magmatism. The Clarence Head dykes represent a large swarm tangential to, and cogenetic with, a giant radiating dyke swarm ~800 km from the indicated source. The preferred mechanism for the emplacement of the Clarence Head dykes is the exploitation of concentric zones of extension around a depleting and collapsing plume source. While the paleomagnetism of most Clarence Head dykes agrees with that of the Franklin dykes, two dykes have anomalous remanence directions, interpreted to be a chemical remanent magnetization carried by pyrrhotite. The pyrrhotite was likely deposited from fluids mobilized southward from the Devonian Ellesmerian Orogeny to the north that used the interiors of the dykes as conduits and precipitated pyrrhotite en route.

Résumé : L'essaim de dykes Clarence Head, à tendance nord-sud, situé sur les îles Devon et Ellesmere dans le Grand Nord canadien, a une tendance orthogonale à l'essaim Franklin (Néoprotérozoïque) qui l'entoure. Les dykes Clarence Head ont été datés par méthode U–Pb sur de la baddeleyite et ils seraient âgés entre 716 ± 1 et 713 ± 1 Ma, des âges apparemment plus jeunes que la plage d'âge des dykes Franklin, mais à l'intérieur des dates publiées pour ces derniers. Le recul alpha dans la baddeleyite pourrait constituer une explication possible pour les différences d'âge, mais une comparaison des âges U–Pb sur des grains de taille semblable des deux essaims suggère que les distances de recul dans la baddeleyite soient inférieures à celles dans le zircon et que les dykes de Clarence Head constituent en effet un événement plus jeune à l'intérieur de la période de magnétisme Franklin. Les dykes Clarence Head représentent un vaste essaim qui est cogénétique avec, et tangent à, un gigantesque essaim de dykes qui rayonne sur une distance d'environ 800 km de la source indiquée. Le mécanisme préférentiel pour la mise en place des dykes Clarence Head est l'exploitation de zones concentriques d'extension autour d'un panache source qui s'appauvrit et qui s'effondre. Bien que le paléomagnétisme de la plupart des dykes Clarence Head concorde avec celui des dykes Franklin, deux dykes ont des directions rémanentes anomales, interprétées comme une magnétisation chimique rémanente portée par la pyrrhotite. La pyrrhotite a probablement été déposée à partir de fluides mobilisés vers le sud à partir de l'orogène ellesmérien (Dévonien) au nord; ces fluides auraient utilisé l'intérieur des dykes comme conduits et auraient précipité la pyrrhotite en chemin.

[Traduit par la Rédaction]

Introduction

Giant mafic dyke swarms are common worldwide (Ernst et al. 1995), and many have radiating patterns of emplacement that appear to indicate their source at the focus, often considered to be a mantle plume (Ernst and Buchan 2001). Examples of mafic dyke swarms tangential (circumferential) to the source have been found in the caldera region of shield volcanoes (e.g., the Galapagos Islands; Walker 1999), but compared with giant radiating dyke swarms, which can extend for well over 2000 km, these are small-scale intrusions with radii of no more than a few kilometres. While no conclusive examples of dyke swarms circumferential to a mantle plume source have been identified to date, it has been

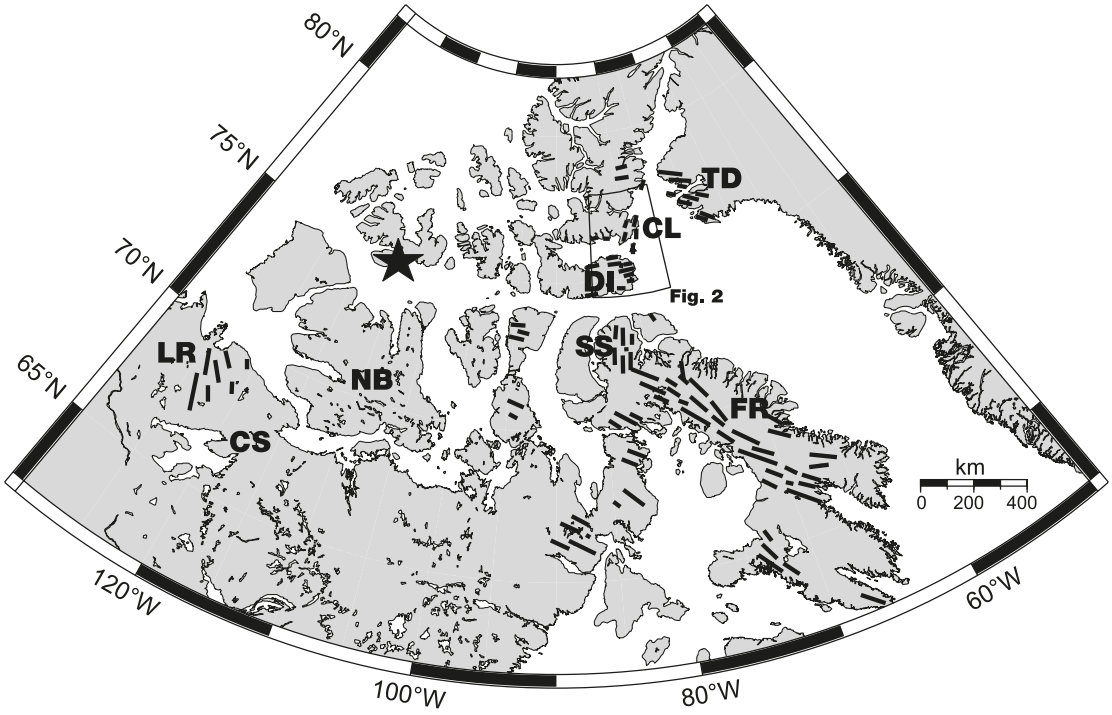
Received 18 November 2008. Accepted 4 March 2009. Published on the NRC Research Press Web site at cjes.nrc.ca on 7 April 2009.

Paper handled by Associate Editor W.J. Davis.

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Fig. 1. Distribution of intrusions or lava flows associated with the Franklin magmatic event, modified and simplified after Fahrig (1987), with dyke locations also from Frisch (1984*a*, 1984*b*, 1984*c*) and Dawes (1991). CL, Clarence Head dykes; CS, Coronation sills; DI, Devon Island dykes; FR, Franklin dykes; LR, Lasard River dykes; NB, Natkusiak basalts (extrusive); SS, Strathcona Sound dykes; TD, Thule dykes. Star denotes approximate location of proposed plume source.



speculated that the Kochikha dykes of Siberia may represent a swarm circumferential to the ca. 250 Ma Siberian Traps plume source and the Blekinge-Dalarna dykes of Scandinavia may be circumferential to the ca. 930 Ma Rogaland massif and related intrusions (Ernst and Buchan 1998), but these associations have not been proven. Extraterrestrial examples may exist in the graben and fissure systems that define the circular structures on Venus (coronae) proposed to have a plume origin (Ernst et al. 2003) and apparent circumferential dyke swarms around volcanic edifices on Mars (Montési 2001), both with radii of between 100–300 km.

The Franklin igneous event emplaced sills and dykes across more than 2500 km of lateral distance in the Canadian Arctic and Greenland (Fig. 1), making it one of the largest dyke swarms on Earth. It comprises large (≥20 m in width) diabase dykes in mainland Canada, the Arctic Archipelago, and Greenland that form a radiating pattern from a proposed plume source to the north of Victoria Island (star on Fig. 1). Also included are the Natkusiak volcanics and sills (NB) of Victoria Island, the Coronation sills (CS) of mainland Northwest Territories, and the Thule sills of Greenland (Denyszyn et al. in press; Shellnutt et al. 2004; Buchan and Ernst 2004; Dawes 1991; Frisch 1984*a*, 1984*b*, 1984*c*; Fahrig et al. 1971).

Southwest of the bulk of the Franklin intrusions, the SSE-trending "Lasard River" diabase dykes (LR in Fig. 1) define a minor swarm, extending over a strike-length of roughly 300 km near the Arctic coast of the Northwest Territories. On the basis of an Ar–Ar (hornblende) age of 708 ± 4 Ma (Ernst et al. 2004), and their paleomagnetism (Park 1981), these dykes (and minor sills) are suggested to be related to the Franklin magmatic event (Buchan and Ernst 2006). Their age and orientation suggest that they could, therefore, be a set of dykes tangential to the Franklin plume and younger than the radial swarm. Another mafic dyke swarm that was tangentially emplaced relative to the proposed Franklin plume is the undated Strathcona Sound swarm (SS in Fig. 1), which has been observed to crosscut the Franklin dykes but has a similar paleomagnetic direction (Christie and Fahrig 1983). While their north-northwest orientation is not quite orthogonal to the trend of the dykes on Baffin Island, which have a southeast–northwest trend, they are located in the NNW-trending Borden fault zone (Jackson et al. 1978), which may have influenced their orientation (e.g., Tokarski 1990). Finally, north–south-trending "Clarence Head" dykes (Buchan and Ernst 2004) are found on Devon Island and Ellesmere Island (CL in Fig. 1). Only preliminary work has previously been done on the Clarence Head swarm (Denyszyn et al. 2006), which has not been observed to crosscut the locally east–west-trending Franklin dykes. There are, therefore, several possible candidates for identifying large-scale emplacement of mafic intrusions tangential to the Franklin plume source.

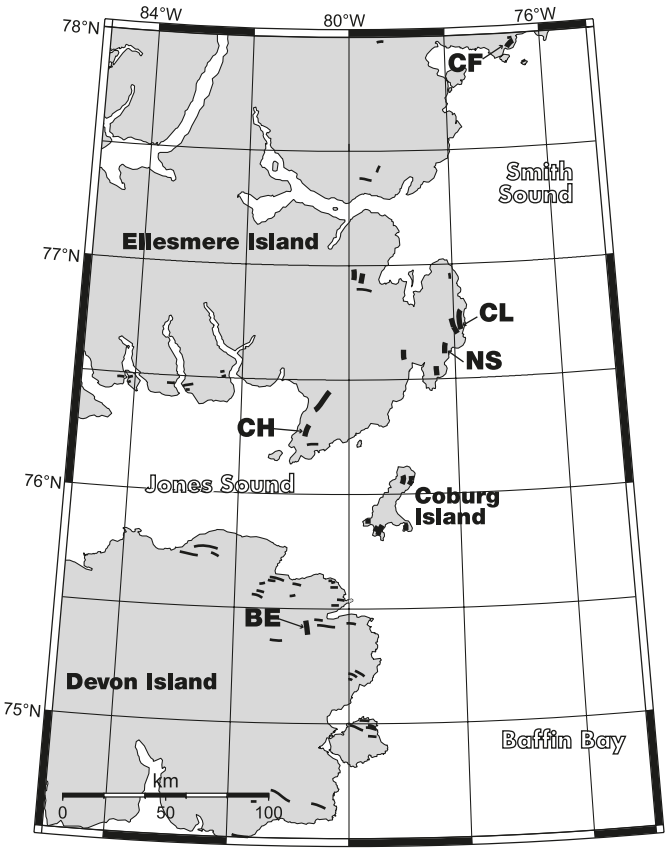
The Franklin intrusions were previously dated by the U/Pb (baddeleyite) method at several locations covering the known extent of the Franklin event, from Victoria Island in the west to the eastern tip of Baffin Island (723^{+4}_{-2} Ma (Heaman et al. 1992) and 720 ± 8 and 716^{+4}_{-5} Ma (Pehrsson and Buchan 1999)). A key paleomagnetic pole (Buchan and Halls 1990) was obtained for the Franklin dyke swarm of 8.4°N, 163.8°E, A_{95} = 2.8°, with a U–Pb (baddeleyite) age range of 721–712 Ma (Denyszyn et al. 2009). This study

presents new paleomagnetic and U–Pb geochronological results for the Clarence Head dykes with an analysis of some complications in the data, as well as a preliminary survey of the effects of alpha recoil in the mineral baddeleyite, commonly used in U–Pb geochronological analyses of mafic rocks.

Geological setting

On Devon Island and southern Ellesmere Island, the basement rocks that host the dykes are high-grade granitic gneiss of Archean to Paleoproterozoic age overlain by flat-lying Paleozoic carbonate and mudstone that also overlie the Franklin dykes (Frisch 1988). The Clarence Head dykes are diabase with abundant clinopyroxene and lath-shaped plagioclase, minor biotite, hornblende, and opaque minerals, with rare titanite and no olivine. Quartz is usually absent, though the Cape Faraday dyke (CF in Fig. 2) did contain some quartz–feldspar microgranophyre. The dykes are near vertical, between 15 and 120 m wide (see Table 1) with well-preserved chilled margins and appear generally unaltered, except for moderate sericitization of plagioclase and occasional chlorite. Dykes CL and NS (Clarence Head and Norton Shaw dykes in Fig. 2) are more fine-grained, with more abundant sulphides and alteration-related phases present relative to other dykes of the Clarence Head swarm. The structural trend of the basement varies locally, but is generally east–west trending on Devon Island and north–south trending on Coburg and Ellesmere islands (Frisch 1984*a*, 1984*b*, 1984*c*). As both the east–west-trending Franklin dyke swarm and the north–south-trending Clarence Head swarm are found throughout the study area regardless of local trends of gneissosity, the orientation of the dykes appears independent of host rock fabric.

Fig. 2. Map showing the locations of dykes included in the study. Heavy lines indicate Neoproterozoic dyke outcrop; Dykes of the Clarence Head swarm are in heavier weight. Dykes: BE, Belcher Glacier; CF, Cape Faraday; CH, Craig Harbour; CL, Clarence Head; NS, Norton Shaw.



Methodology

Five north–south-trending dykes from Devon and Ellesmere Island were sampled (Fig. 2), with samples taken from each for paleomagnetic analysis. For geochronological analyses, block samples were taken from the coarser grained interiors of the dykes, as these are considered more likely to contain baddeleyite grains large enough to analyze as single grains (e.g., Heaman and LeCheminant 1993).

Paleomagnetism

Samples were collected either as field-drilled oriented core or as oriented block samples. A sun compass was used for orientation of the samples, though uneven terrain or weather conditions made sighting the sun impossible at sites BE and NS. In these cases, multiple checks on the reliability of the magnetic compasses (such as retaking the compass reading away from the possibly magnetic dyke rocks) were made before accepting a reading as satisfactory. All sites were located by onboard helicopter GPS (global positioning system).

At least one cylindrical specimen 2.45 cm in length and diameter from each sample was subjected to detailed alternating field (AF) demagnetization, using a Schonstedt GSD-1 instrument. Field increments averaged 5 mT up to maximum fields of 95 mT. After each demagnetization step, the direction and intensity of remanent magnetization were measured using a modified Digico spinner magnetometer with reproducibility to magnetization intensities as low as $\sim 10^{-3}$ A m^{-1}. Measurement procedure included an averaging algorithm (Halls 1986) to reduce the dependence of results on the last sample axis to be demagnetized in the single-axis demagnetizer. Thermal demagnetization was also used for selected specimens, using a Schonstedt TSD-1 unit. Stepwise heating was conducted according to a measurement routine that involved taking measurements at temperature increments that decreased as the unblocking temperature of magnetite (580 °C) was approached. The directional data were analyzed using stereoplots and vector diagrams (Zijderveld 1967), as well as principal component analysis (Kirschvink 1980) that included a search routine to find all linear segments on vector diagrams consisting of three or more points that pass a minimum acceptance criteria given by a maximum angle of deviation (Kirschvink 1980), which was set at ≤10°. Susceptibility versus temperature data were obtained from one or two specimens from every dyke using a Sapphire Instruments SI3 meter to determine remanence carriers. A more detailed analysis of specimens from the NS and CL dykes was undertaken as the paleomagnetic results indicated a remanence carrier with a low unblocking temperature (~300 °C) in addition to Ti-poor magnetite. During measurement, the specimens were heated from 20 to 650 °C, and then cooled. Susceptibility measurements automatically re-

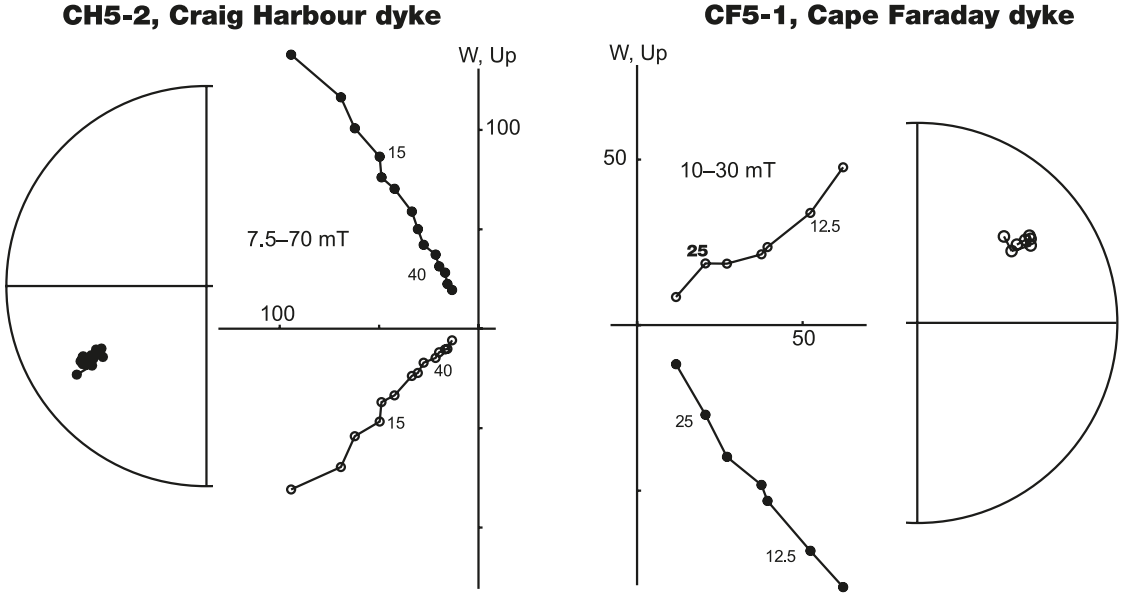
Table 1. Summary of paleomagnetic results.

Site	Latitude (N)	Longitude (W)	Str (°)	*W* (m)	N_S	*D* (°)	*I* (°)	P_{lat} (°N)	P_{lon} (°E)	*k*	α_{95}
Clarence Head dykes											
CF (R)	77°53.9	76°38.8	38	120	3	58.2	–35.6	12.8	228.6	75	5.2
CL	76°42.9	77°50.9	355	25	6	165.7	10.3	–8.0	296.5	75	7.8
NS* (cm)	76°39.7	78°04.9	350	20	4	133.7	32.5	8.1	326.1	105	9.1
NS* (int)	76°39.7	78°04.9	350	20	4	171.3	24.0	–0.3	290.5	40	14.6
CH	76°18.3	78°04.9	20	60	4	240.2	32.4	10.4	222.8	59	10.4
BE* (R)	75°34.2	81°25.0	355	15	3	107.5	–29.6	19.7	176.1	40	19.8

Note: *W*, intrusion width; N_S, number of samples; Str, strike direction; *D*, mean declination; *I*, mean inclination; P_{lat}, paleolatitude; P_{lon}, paleolongitude; *k*, Fisher precision parameter; α_{95}, radius of 95% confidence circle about the mean; (R) denotes reversed-polarity dyke. BE, Belcher Glacier; CF, Cape Faraday; CH, Craig Harbour; CL, Clarence Head; NS, Norton Shaw.

*denotes samples oriented using only a magnetic compass. (cm) denotes chilled margin; (int) denotes interior.

Fig. 3. Examples of stable end-points obtained from dykes of the Clarence Head swarm. On equal-area stereoplots solid and open symbols are downward and upward-pointing magnetizations, respectively. On vector diagrams solid circles are projections of the tip of the magnetization vector on the horizontal plane, and open circles are projections on the vertical plane.



corded at 5 °C intervals during both heating and cooling stages.

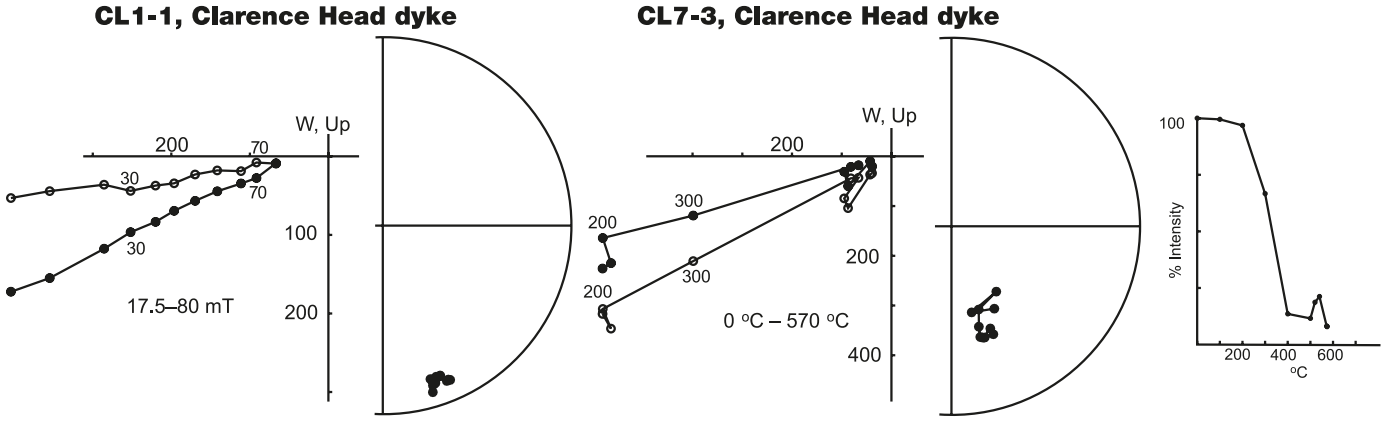
U–Pb geochronology

The extraction of baddeleyite (ZrO_2) was attempted for geochronological analysis. The density-based mineral separation methods used here would also include zircon in the final concentrate, though zircon was not found to be present in any of the dykes. Meticulous cleaning practices were followed at every stage to minimize any chance of sample contamination. As mineral-separation techniques typically used for zircon (i.e., using a Wilfley table, Frantz magnetic separator, and heavy liquids) proved inadequate for isolating baddeleyite, alternate methods were needed. Sample CH was crushed and milled, and the powder was then repeatedly stirred and allowed to settle in water, decanting off the suspended fine fraction. A handheld magnet was used to extract magnetic material (magnetite and (or) iron from the disc mill) from the sediment, which was subsequently put through a heavy liquid separation using methylene iodide. The heavy fraction was then examined and handpicked under a binocular microscope. This yielded several fresh baddeleyite crystals and fragments, while another sample from the same dyke had failed to yield any baddeleyite using conventional procedures.

The Wilfley water-shaking table was used for subsequent samples by a technique modified after Söderlund and Johansson (2002) that involved adding a teaspoonful of the powdered sample to the Wilfley table at a time and collecting a residue of the highest density crystals directly off the surface of the table with a pipette, followed by handpicking baddeleyite grains under ethanol using a binocular microscope.

Baddeleyite crystals are generally too small and fragile to be abraded, but shiny surfaces and crisp edges on the baddeleyite crystals chosen for analysis indicated that they were unaltered. Dissolution, isotope dilution, and sample loading methods were as described in Krogh (1973), using a ^{205}Pb–^{235}U spike and miniature bombs. This spike has been calibrated against several artificial Pb–U mixtures prepared in-house and obtained through the EARTHTIME project. The Pb/U values determined from these measurements agree to well within 0.1%. No chemical separation

Fig. 4. Examples of stable paleomagnetic end-points from the Clarence Head dyke (CL), showing the "anomalous" paleomagnetic direction and the low unblocking temperature, high-coercivity component. Symbols as in Fig. 3. Thermal data include an additional intensity decay plot.



procedures were required because of the small size of the samples. The Pb and U isotopes were analyzed on a VG354 thermal ionization mass spectrometer (TIMS) using a Daly collector in pulse counting mode. The mass discrimination correction for this detector is constant at 0.07%/AMU (atomic mass units). Thermal mass discrimination corrections are 0.10%/AMU. Dead time of the measuring system was about 20 ns and was monitored using the SRM982 standard.

Results

Paleomagnetism

Three dykes with a north–south trend produced identical paleomagnetic results to those of the radial Franklin dykes (Denyszyn et al. 2009), one (CH) with a west-directed remanence direction, and two (BE, CF) with a direction roughly antipodal (Table 1, Fig. 3). Two of the Clarence Head dykes (CL and NS) have an anomalous direction compared with the "Franklin" direction of the other three (Fig. 4). Both of these characteristic components have typically high coercivity, but that of CL and NS has a distinctly lower unblocking temperature (ca. 320 °C) though the specimens often maintain a high intensity of magnetization after AF cleaning to almost 100 mT.

Temperature versus susceptibility plots (Fig. 5) show a sharp drop at about 580 °C for the sites with the "Franklin" paleomagnetic direction (Fig. 5*a*), indicating Ti-poor magnetite as the carrier of the characteristic magnetization in these dykes; but notable in Fig. 5*b* is the presence of a pronounced asymmetric peak in susceptibility for samples from site CL (with an "anomalous" remanence direction) in the heating run before a sharp decrease at 320 °C. The bulk susceptibility (K_0) is also an order of magnitude lower in site CL than in those with magnetite as the principal remanence carrier. A small drop in susceptibility at 580 °C indicates the presence of magnetite, but the low bulk susceptibility value suggests only a small contribution. All samples from dyke CL contain the remanence with the southerly declination, but none from dykes CH, BE, or CF carry it. The NS dyke shows a variation in magnetization direction from the margins of the dyke to the interior (Table 2, Fig. 6). The fine-grained specimens from the margin (Fig. 6*a*) have a remanence direction charac-

Fig. 5. Example plots of susceptibility (K) at temperature (in °C), normalized to room temperature susceptibility (K(0)), for (*a*) representative dykes from the Clarence Head swarm that have the "typical Franklin" remanence direction and (*b*) site CL, a dyke in the Clarence Head swarm with an anomalous remanence direction. Note the lower bulk susceptibility in (*b*) and the asymmetric peak at ~300 °C.

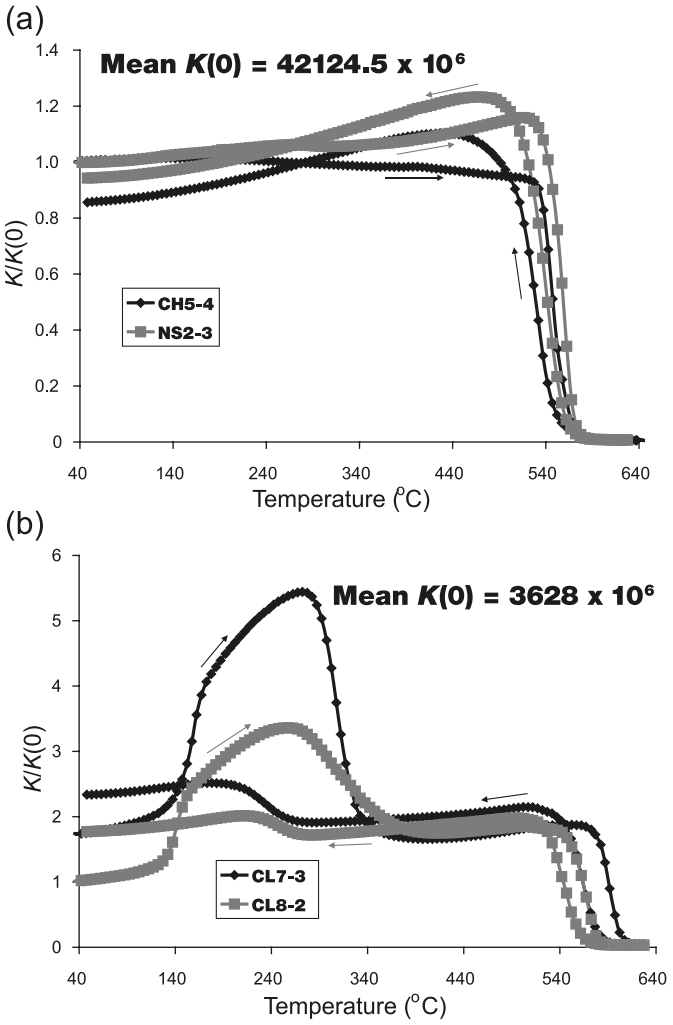


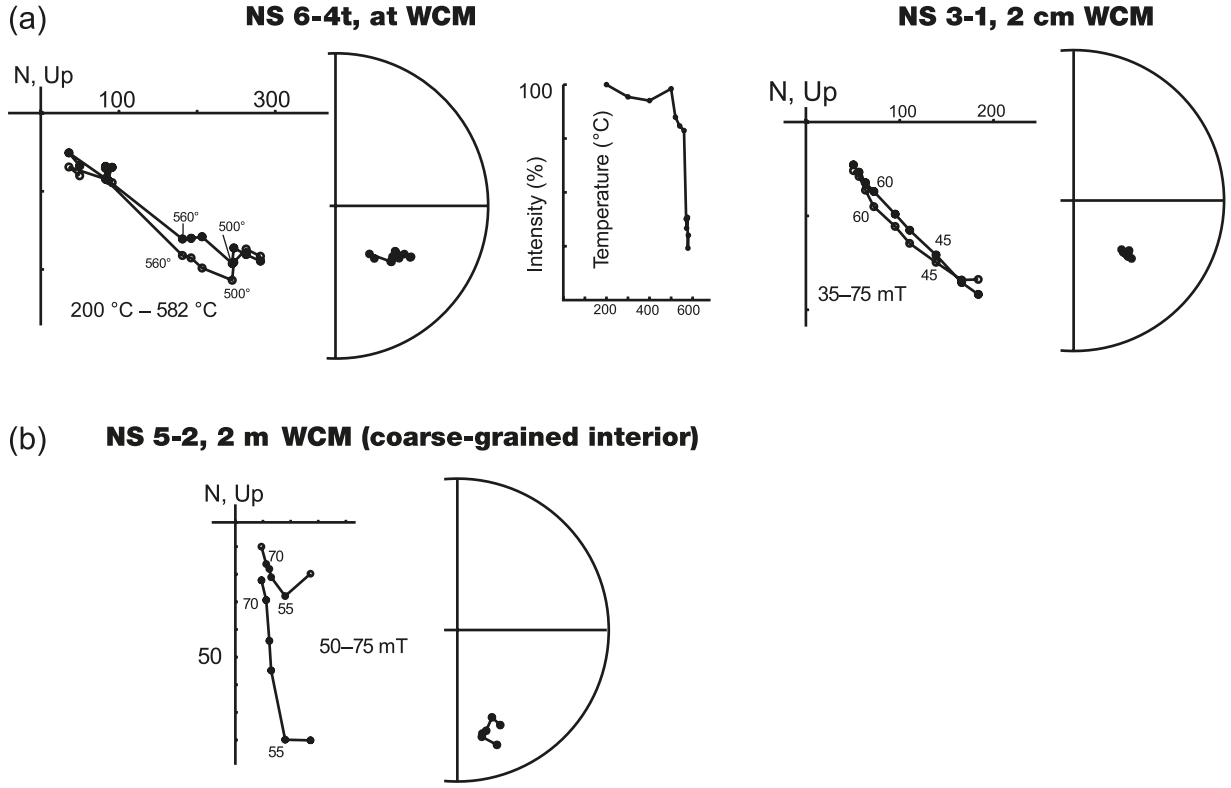
Table 2. U–Pb isotopic data for baddeleyite from mafic dykes in Canada and Greenland.

Sample	Fraction	w_{mod} (mg)	d_{mod} (μm)	Th/U	cPb (pg)	$^{206}Pb/^{204}Pb$ (measured)	$^{206}Pb/^{238}U$	2σ	$^{207}Pb/^{235}U$	2σ	$^{206}Pb/^{238}U$ age (Ma)	2σ	$^{207}Pb/^{206}Pb$ age (Ma)	2σ	% (disc.)	ρ conc.
CF-1, Cape Faraday diabase, Ellesmere Island																
1	2 frags	0.0004	8.5	0.001	4.0	41.2	0.117480	0.004	1.04898	0.436	716.0	22.2	766.5	950.4	7.0	0.967
2	2 frags	0.0001	16.9	0.005	1.4	209.1	0.116944	0.001	1.01791	0.052	712.9	3.4	712.6	103.0	–0.1	0.796
3	7 fresh, pale blade frags	0.0001	4.4	0.002	3.1	93.4	0.116868	0.001	1.03334	0.135	712.5	7.3	745.8	266.6	4.7	0.922
CH-1, Craig Harbour diabase, Ellesmere Island																
1	1 cracked frag	0.0004	7.7	0.018	0.6	1428.2	0.117045	0.001	1.02586	0.009	713.5	3.3	727.3	16.1	2.0	0.489
2	1 square frag	0.0001	4.4	0.002	0.6	470.5	0.116925	0.001	1.02461	0.033	712.8	3.3	726.8	64.2	2.0	0.487
CL-2, Clarence Head diabase, Ellesmere Island																
1	1 grain	0.0007	41.4	0.014	1.8	906.9	0.117157	0.000	1.02170	0.012	714.2	1.5	716.6	22.0	0.4	0.559
2	1 grain	0.0009	26.2	0.002	1.0	621.2	0.117575	0.000	1.02640	0.019	716.6	2.0	718.8	35.9	0.3	0.591
3	1 grain, tri-angular	0.0005	59.0	0.031	0.3	4023.8	0.117536	0.000	1.02466	0.004	716.4	1.5	715.9	6.2	–0.1	0.690
4	1 blocky frag	0.0002*	18.3*	0.022	0.3	5500.2	0.117718	0.000	1.03054	0.003	717.4	1.4	724.8	4.3	1.1	0.781
5	1 blocky frag	0.0001*	26.7*	0.007	0.3	1524.4	0.116978	0.000	1.01925	0.008	713.1	2.7	714.8	14.2	0.2	0.602
6	1 long blade frag	0.0001*	16.8*	0.003	0.2	3228.4	0.117579	0.000	1.03090	0.005	716.6	1.7	728.0	6.8	1.7	0.698

Note: Errors are given at 2σ. Sample locations are given in Table 1. frag(s), fragment(s); w_{mod}, weight based on images normal to [100] face and assuming 300 ppm U; d_{mod}, thickness based on images normal to [100] face and assuming 300 ppm U; Th/U is calculated from radiogenic $^{208}Pb/^{206}Pb$ ratio and $^{207}Pb/^{206}Pb$ age assuming concordance; cPb, total measured common Pb assuming the isotopic composition of laboratory blank: 206/204 = 18.221; 207/204 = 15.612; 208/204 = 39.360 (errors of 2%); disc., % discordance for the given age; ρ conc., error correlation coefficient.

*denotes weights/thicknesses determined by actual volume measurements based on scanning electron microscopy (SEM) imaging.

Fig. 6. Examples of stable paleomagnetic end-points from the Norton Shaw dyke (NS), showing (*a*) the "Franklin" paleomagnetic direction and the unblocking temperature indicative of magnetite being the remanence carrier in samples from the fine-grained chilled margin, and (*b*) the "anomalous," high-coercivity component in specimens from the coarser grained interior. WCM, west chilled margin. Symbols as in Fig. 3.



teristic of the Franklin direction, with unblocking temperatures characteristic of magnetite (i.e., ca. 580 °C), while coarser grained specimens from the interior (Fig. 6*b*) yield the more southerly direction, though thermal demagnetization did not reveal the presence of significant amounts of pyrrhotite in samples with the anomalous direction.

U–Pb geochronology

The uranium and lead isotopic results for each analysis are given in Table 2 and plotted in Fig. 7, with all errors and error ellipses at 2σ. The data were corrected for common lead contamination using the isotopic composition of laboratory blank (see footnotes to Table 2), and total common lead is typically <1 pg. Even so, because of the small size of the fractions (as low as 0.1 μg), a result of the tiny size of even the best quality crystals, a significant proportion of the total ^{207}Pb is common. The low laboratory blanks and concordant data nevertheless permit precise ages to be obtained from the ^{238}U–^{206}Pb system, and the ages presented here are weighted averages of the ^{238}U–^{206}Pb ages of the fractions from each sample. Averages of ^{206}Pb/^{238}U ages and fit parameters were calculated and plotted using programs of Davis (1982) and Ludwig (2003), and U decay constants are from Jaffey et al. (1971).

The north–south-trending dykes of the Clarence Head swarm (Fig. 7) appear to be significantly younger than the Franklin dykes. The dyke at Cape Faraday (CF) was dated with three analyses on fractions of two to seven crystals each. The baddeleyite recovered from this sample were pale, very thin, fresh blades between 20–50 μm long. The data overlap within error and yield an average ^{206}Pb/^{238}U age of 713 ± 3 Ma (Fig. 7*a*).

The Craig Harbour (CH) dyke yielded several relatively large (>100 μm) grains when use of modified mineral separation techniques were employed, after those used to isolate zircon had failed (see "Methodology" section). The honey-brown striated blades and fragments were thin and flat, and were analyzed as single grains. A ^{206}Pb/^{238}U age of 713 ± 2 Ma (Fig. 7*b*) was obtained based on two overlapping, concordant analyses of single baddeleyite crystals.

The Clarence Head (CL) dyke, one of the two north–south-trending dykes with an anomalous paleomagnetic direction, yielded several dark brown, relatively large (~100 μm) blocky grains, as well as thin blade fragments. Six single-crystal fractions yield overlapping data with a slightly older ^{206}Pb/^{238}U age of 716 ± 1 Ma (Fig. 7*c*). The north–south-trending dykes (Fig. 7) appear to be significantly younger than those previously published for the Franklin dykes (Heaman et al. 1992; Pehrsson and Buchan 1999; Denyszyn et al. 2009).

Discussion

Geochronology of baddeleyite in the Franklin intrusions

Because of the small size of the baddeleyite crystals, possible effects of alpha recoil (Kigoshi 1971; Fleischer 1982; Romer 2003) must be considered in these and previously published results. Decay of an alpha-emitting radionuclide

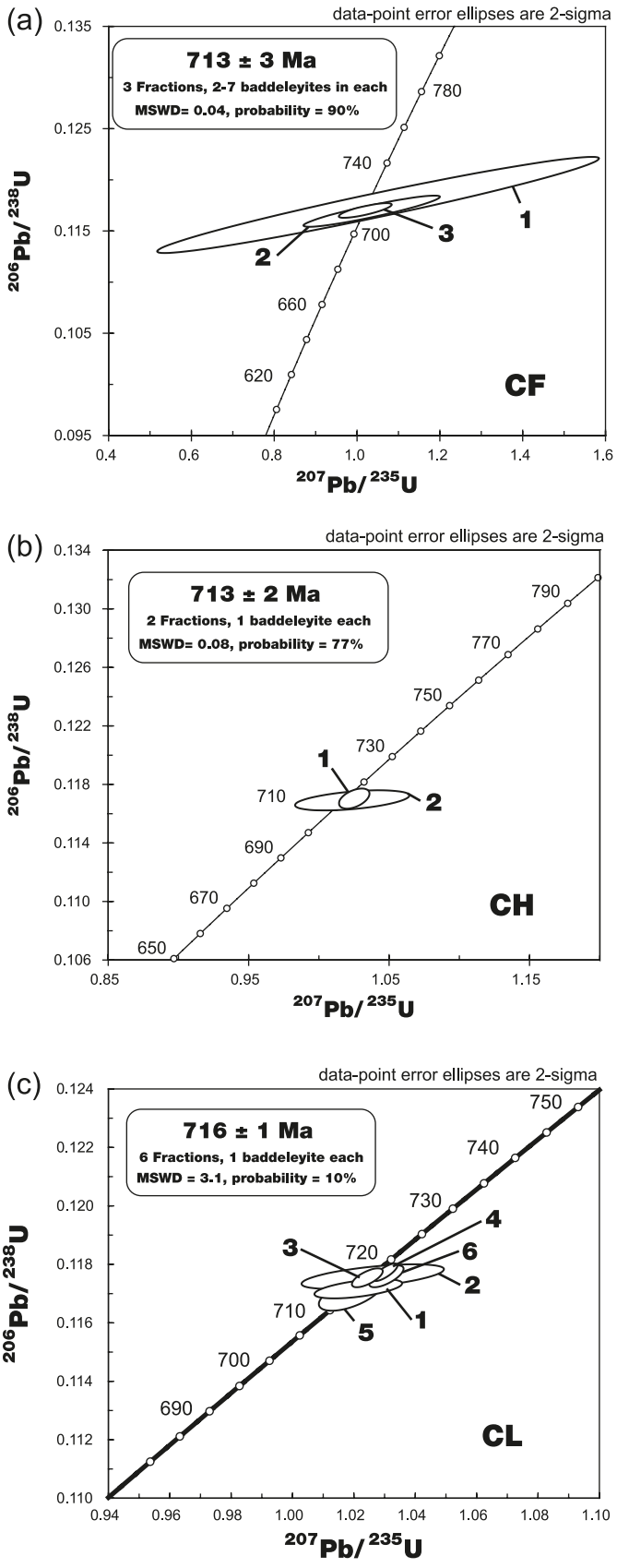


Fig. 7. Concordia plots showing geochronological results for north–south-trending Clarence Head dykes of Ellesmere Island: (*a*) the Cape Faraday (CF) dyke, (*b*) the Craig Harbour (CH) dyke, and (*c*) the Clarence Head (CL) dyke. MSWD, mean square of weighted deviates.

causes the daughter nucleus to recoil through the crystal lattice. For zircon, recoil distances vary from about 20 nm to 30 nm, but there are eight alpha decays in the ^{238}U system and seven in the ^{235}U system, so the average total displacement for a radiogenic Pb isotope is estimated to be roughly 0.1 μm (Romer 2003, and included references). Any Pb loss due to this mechanism will lower the ^{206}Pb/^{238}U age. This is likely to be minor (<0.1%) for most zircon analyzed by TIMS, where the outer surface is removed by laboratory physical or chemical abrasion (Krogh 1982; Mattinson 2005) and all grain dimensions are normally in excess of 50 μm. However, baddeleyite is not abraded and often takes the form of extremely thin tabular crystals, raising the possibility that significant recoil loss could have occurred normal to [100] faces. For recoil distances comparable to those of zircon (average alpha recoil track length of 0.025 μm) and assuming a quasi-infinite tabular geometry, the level of Pb loss would scale inversely with the thickness and should be about 0.45% for a grain thickness of 10 μm.

The 723^{+4}_{-2} Ma age on the Franklin swarm of Heaman et al. (1992) is based on a regression of slightly discordant data with a lower intercept close to 0 Ma. It is, therefore, relatively insensitive to alpha recoil effects, which would produce only second-order biases in ^{207}Pb/^{206}Pb ages. However, ^{206}Pb/^{238}U ages on baddeleyite from east–west-trending dykes on Devon and Ellesmere islands (Denyszyn et al. 2009) gave an average of 721 Ma. The consistency of this age with the ^{207}Pb/^{206}Pb age of Heaman et al. (1992) argues that the older fractions preserve accurate ages. The younger ages presented here from three north–south-trending dykes fall within the range of ^{206}Pb/^{238}U ages observed in the composite data set of Heaman et al. (1992) but are based on fractions with far fewer grains. Due to the large ^{207}Pb/^{235}U errors, their discordance cannot be well evaluated. Each of these structurally distinct dykes, where analyses were mostly done on fractions of 1–2 grains so there is a minimal averaging effect, gives consistent overlapping ^{206}Pb/^{238}U ages. However, the possibility must be addressed that this relatively young age could be an artifact of alpha recoil.

Crystals dated in the early part of the present study, before we realized the potential importance of alpha recoil effects, were only photographed normal to the wide [100] faces using an optical microscope. Examination of similar-size grains from the same samples indicates thicknesses that vary in the approximate range 6 to 20 μm. Therefore, reduction in apparent age of about 5 million years might be expected for the thinner grains. Model thicknesses (d_{mod}) for these analyzed grains are given in Table 2, based on an assumed U concentration of 300 ppm. U concentrations in most published analyses of large, accurately weighed baddeleyite fractions average about 300 ppm and do not vary much more than 50% about this value (e.g., Krogh et al. 1987). Calculated thicknesses based on this assumption show no consistent correlation with ^{206}Pb/^{238}U ages. As a further check on potential alpha recoil effects, grains from the east–west-trending (Franklin) CG-7 dyke of Denyszyn et al. (in press) (Fig. 2) and the north–south-trending CL-1 dyke were imaged at a high angle using a scanning electron

Fig. 8. Scanning electron microscopy (SEM) photomicrographs of single baddeleyite crystals taken to establish crystal dimensions and more accurate weights. Shown are images of examples from (*a*) the Cadogan Glacier (CG) dyke; (*b*) the Clarence Head (CL) dyke. Scale bar in each is 20 μm.

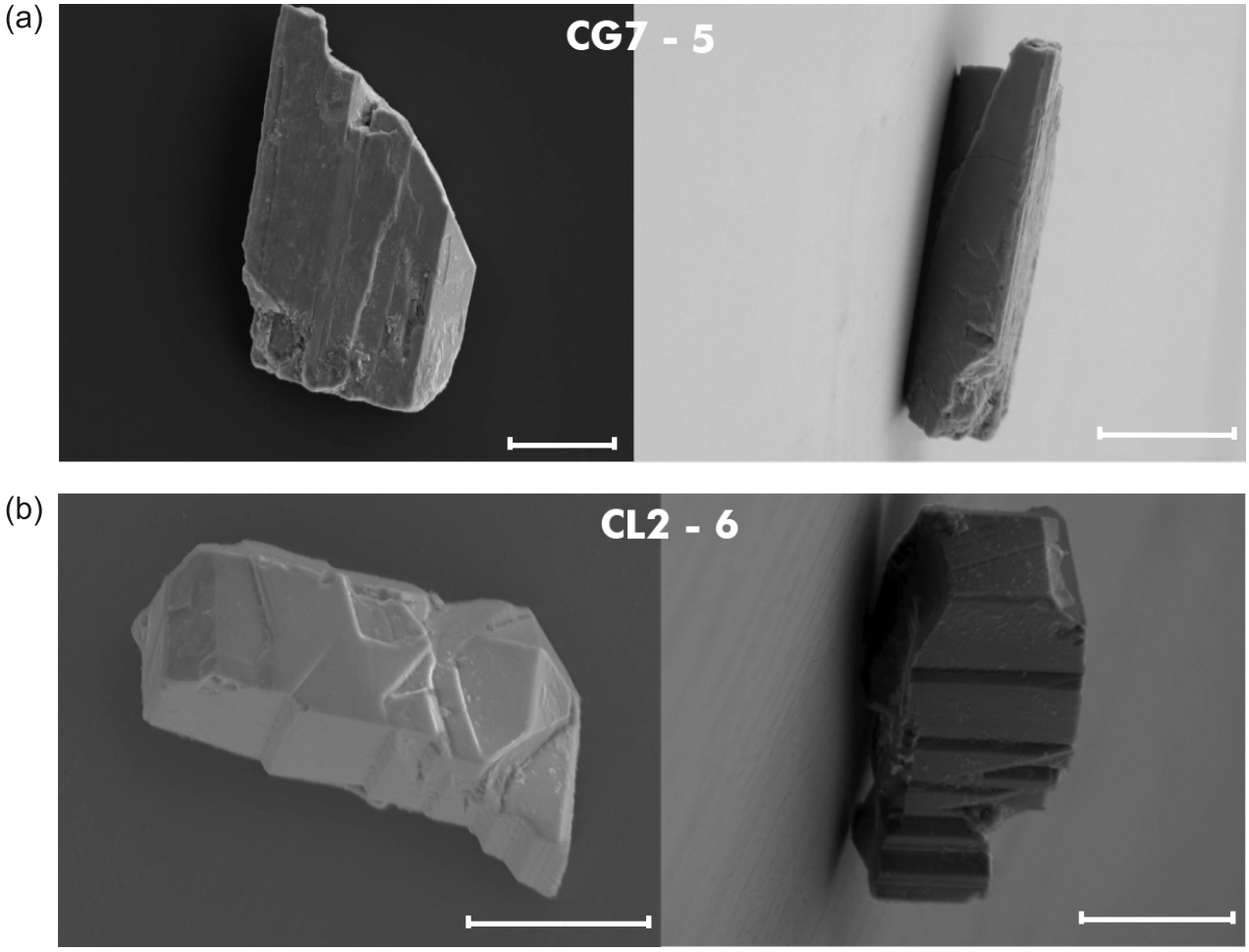
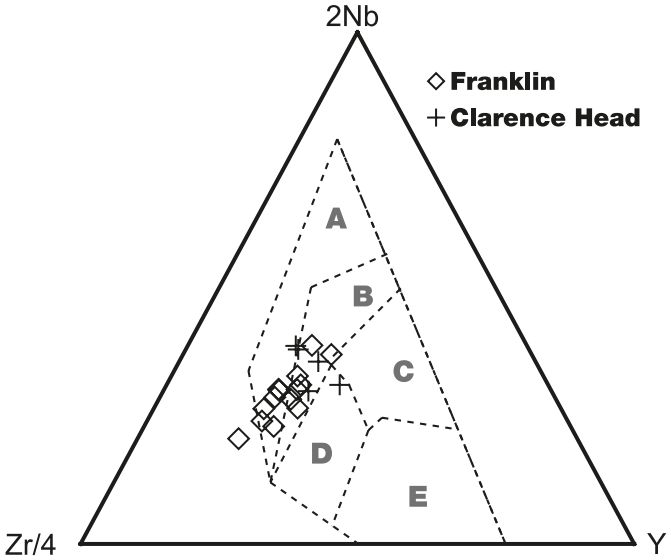


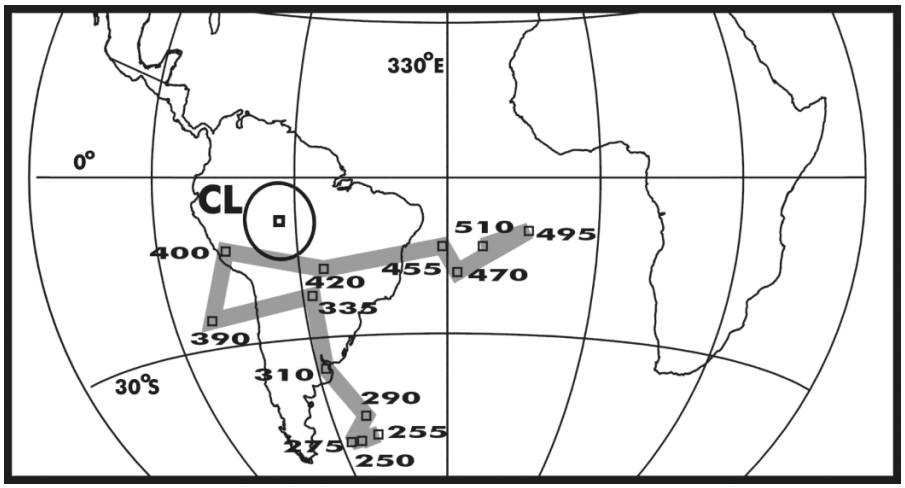
Fig. 9. 2Nb–Zr/4–Y plot (Meschede 1986) of whole-rock analyses of the Clarence Head dykes and Franklin dykes from Devon and Ellesmere islands (data from Denyszyn 2008). Lettered fields denote tectonic setting: A and B, within-plate alkaline; B and D, within-plate tholeiite; C, primitive mid-ocean ridge basalt (P-MORB); E, normal MORB (N-MORB); D and E, volcanic arc basalt.



microscope (SEM) to determine precise dimensions, including thickness, before analysis (Fig. 8). The SEM images also show the crystals to be pristine, without any indication of microscale breakdown of baddeleyite to zircon that could result in minor Pb loss and discordance. The results from two analyses from the Cadogan Glacier dyke of the Franklin swarm (Denyszyn et al. 2009) and analyses 4, 5, and 6 from the CL-1 dyke indicate that there is no such correlation for thicknesses varying over the range 10–20 μm. While the U concentrations for the accurately measured grains from the CG-7 dyke were close to the "model" value of 300 ppm U (CG7-5: 395 ppm, CG7-6: 371 ppm), those from the CL-2 dyke have significantly higher U concentrations (CL2-4: 1127 ppm, CL2-5: 623 ppm, CL2-6: 929 ppm). These observations argue that the north–south-trending dykes form a distinct younger age group. An additional conclusion is that alpha recoil distances are either significantly smaller in baddeleyite than in zircon or that previous estimates based of recoil distances in zircon are too high.

Previously published ages for the Franklin swarm were determined exclusively from dykes that are part of the radial

Fig. 10. Apparent polar wander path for North America (from Torsvik et al. 1996) and the virtual geomagnetic pole for site CL.



pattern of the swarm, while the Clarence Head dykes are tangential to the focus of the swarm. Other sets of tangential dykes, the Strathcona Sound dykes on north Baffin Island and the Lasard River dykes of the mainland Northwest Territories (Buchan and Ernst 2004), have not been dated although the paleomagnetism of each (Christie and Fahrig 1983, Park 1981) indicates an affiliation with the Franklin event. The observation that the Strathcona Sound swarm crosscuts the radial Franklin dykes in at least two locations (Christie and Fahrig 1983) further supports the conclusion that the tangential dykes represent a distinct, younger event. The Clarence Head dykes are also indistinguishable geochemically from nearby Franklin dykes (Fig. 9), an observation that supports a shared affiliation with the source of the radial swarm.

Emplacement of orthogonal dykes

While the more common radial pattern of dyke swarms is a result of the uplift of the lithosphere by the impinging mantle plume and injection of dykes from an overpressured magma reservoir (e.g., Baragar et al. 1996), concentric zones of extension may form due to the depression of this source area as the plume reservoir is depleted and concentric zones of compression during lithospheric uplift undergo gravitational relaxation (e.g., McKenzie et al. 1992; Mège and Ernst 2001). Magma could have been continuously or episodically transported along pre-existing conduits at a depth of neutral buoyancy (Lister and Kerr 1990) before emplacement in the upper crust according to the prevailing stress regime. Because the north–south-trending dykes are younger, it is possible that they were emplaced orthogonal to the east–west-trending dykes of the main Franklin swarm because of extension caused by the plume head depleting and collapsing.

Another explanation for orthogonality of virtually coeval dykes is a change in the regional deviatoric stress field as a result of emplacement of the east–west-trending dykes to allow the emplacement of the north–south-trending dykes (radial to tangential; e.g., Féraud et al. 1987). If the regional horizontal stress field was relatively low and uniform such that the intrusion of the radial Franklin dykes changed the maximum principal stress direction by 90° (i.e., σ_1 became σ_3), this could have allowed the emplacement of the Clarence Head and possibly the Lasard River and Strathcona Sound swarms. The absence of any significant change in composition supports the interpretation that the dykes, in either orientation, are being fed from the same source.

Paleomagnetic evidence for large-scale fluid movement in the Devonian

The most significant difference between the radial and tangential Franklin dykes is the anomalous remanence direction of dykes CL and NS, which also feature the prominent asymmetric peak in susceptibility versus temperature plots (Figs. 4, 5). Previous studies (e.g., Halls et al. 2001) attribute the peak to the presence of the Fe_9S_{10} form of pyrrhotite, which exhibits ferrimagnetic properties in the vicinity of 200 °C when antiferrimagnetic hexagonal pyrrhotite undergoes a crystallographic transformation to a ferrimagnetic phase; then at ~230 °C, another transformation to antiferrimagnetic monoclinic pyrrhotite occurs (Dekkers 1989). The lack of a significant increase in susceptibility at 320 °C on the cooling run indicates that the pyrrhotite has been oxidized to magnetite, a transformation that typically occurs at temperatures higher than 500 °C (McElhinny and McFadden 2000), an interpretation supported by the significant increase of susceptibility on the cooling run. Another mineral that also displays this behaviour is lepidocrocite, an Fe-oxyhydroxide that can convert to maghaemite upon heating (Schmidt 1993). However, lepidocrocite tends to be a red-coloured precipitate from extensive tropical weathering and soil-forming processes, and the dyke rock showed no external signs of weathering, nor of any red colour. Therefore, the anomalous remanence direction is interpreted as a chemical remanent magnetization (CRM) carried by pyrrhotite, which was precipitated by fluids expelled during the accretion of the Pearya terrane to the north of Ellesmere Island in the Devonian (Trettin 1991). These fluids were also responsible for the formation of the Polaris Mississippi Valley-type ore deposit on Little Cornwallis Island, due west of Devon Island (Symons and Sangster 1992). This is supported by the position of the virtual geomagnetic pole (VGP) of the CL dyke relative to the North American apparent polar wander path (Fig. 10), where it fits reasonably well to the Late Devonian – Early Silurian part of the appa-

rent polar wander path (APWP), corresponding to the time of accretion of Pearya. The north–south orientation of the dykes probably made them effective fluid conduits (Denyszyn et al. 2006), and their petrography also indicates a much greater degree of alteration compared to other Franklin dykes. Another line of evidence supporting the fluid-flow hypothesis is the observation of the difference in remanence directions measured in the samples from the interior of the NS dyke versus those from its chilled margin (Fig. 6). Samples from the interior tend to have the more southerly direction characteristic of the CL dyke (declination ca. 170), where it is carried by pyrrhotite, while samples from the margin have a more easterly declination (ca. 135), characteristic of the Franklin direction. The coarser grained interior of the dyke would have permitted more fluid flow and allow the precipitation of pyrrhotite, and the permeability of the interior may have also been enhanced by the presence of dyke-parallel joints. While shearing often occurs along dyke margins and, therefore, might make them channels for fluid flow, thin section analysis does not show any evidence for shearing or fracturing that would increase permeability in this case.

This observation also places limits on the maximum temperature to which the rocks have been heated since their formation in the Neoproterozoic. Apatite fission-track and (U–Th–Sm)/He thermochronological studies, capable of resolving thermal events through temperatures of 40–130 °C, were carried out in the region (Grist and Zentilli 2005), revealing a spread of fission-track ages more or less continuous from the Late Carboniferous through the Paleocene, related to periods of erosion and clastic sedimentation at various times throughout the Canadian Arctic Archipelago. This study shows that for southeast Ellesmere Island at least, at no time since the Devonian did temperatures of the rocks exceed 320 °C.

Conclusions

This study represents the first documentation of cogenetic dykes orthogonal to a giant radiating dyke swarm, remarkable for having been emplaced ~800 km from the proposed plume source. Baddeleyite fractions from three dykes from the north–south-trending Clarence Head swarm give ages between 716 ± 1 and 713 ± 1 Ma. The consistency of results on grains with estimated thickness ca. 10 μm suggests that alpha recoil distances are significantly lower in baddeleyite than they appear to be in zircon and that the Clarence Head swarm represents a distinctly younger event within the period of Franklin magmatism. The Strathcona Sound dykes are also likely a part of this event. The north–south-trending mode of emplacement may have resulted from a change in the regional stress regime, due either to the collapse of a continually depleting plume head source or modifications to the regional stress pattern during emplacement of the earlier radial Franklin swarm. Paleomagnetic results for two Clarence Head dykes reveal the effects of chemical remagnetization, likely a result of the precipitation of pyrrhotite as fluids mobilized by the Ellesmerian Orogeny in the Devonian used the coarse-grained interiors of the dykes as conduits.

Acknowledgments

Kim Kwok and Mike Hamilton of the Jack Satterly Geochronology Laboratory at the University of Toronto are thanked for help with geochronological analyses. Funding was provided by a Natural Sciences and Engineering Research Council of Canada Discovery Grant to HCH and a Northern Scientific Training Program grant to SWD. The Polar Continental Shelf Project is gratefully acknowledged for providing logistical support for field work in the Canadian Arctic (2002–2005). Paleomagnetic work was carried out at the Paleomagnetism Laboratory at the University of Toronto, Mississauga. M. Wingate, U. Söderlund, and W.J. Davis are thanked for useful comments and reviews.

References

Baragar, W.R.A., Ernst, R.E., Hulbert, L., and Peterson, T. 1996. Longitudinal petrochemical variation in the Mackenzie dyke swarm, northwestern Canadian Shield. Journal of Petrology, **37**: 317–359. doi:10.1093/petrology/37.2.317.

Buchan, K.L., and Ernst, R.E. 2004. Dyke swarms and related units in Canada and adjacent regions scale 1 : 5 000 000. Geological Survey of Canada, Map 2022A, with notes.

Buchan, K.L., and Ernst, R.E. 2006. Giant dyke swarms and the reconstruction of the Canadian Arctic islands, Greenland, Svalbard, and Franz Josef Land. *In* Dyke swarms—time markers of crustal evolution. *Edited by* E. Hanski et al. Taylor & Francis, London, UK., pp. 27–48.

Buchan, K.L., and Halls, H.C. 1990. Paleomagnetism of Proterozoic mafic dyke swarms of the Canadian Shield, *In* Mafic dyke swarms and emplacement mechanisms. *Edited by* A. J. Parker et al. A.A. Balkema, Rotterdam, The Netherlands, pp. 209–230.

Christie, K.W., and Fahrig, W.F. 1983. Paleomagnetism of the Borden dykes of Baffin Island and its bearing on the Grenville Loop. Canadian Journal of Earth Sciences, **20**: 275–289.

Davis, D.W. 1982. Optimum linear regression and error estimation applied to U–Pb data. Canadian Journal of Earth Sciences, **19**: 2141–2149. doi:10.1139/e82-019.

Dawes, P.R. 1991. Geological compilation map of the Thule area, Greenland. Geological Survey of Greenland, Publication, scale 1 : 500 000.

Dekkers, M.J. 1989. Magnetic properties of natural pyrrhotite II: High and low temperature behaviour of J_{IS} and TRM as a function of grain size. Physics of the Earth and Planetary Interiors, **57**: 266–283. doi:10.1016/0031-9201(89)90116-7.

Denyszyn, S.W. 2008. Paleomagnetism, geochemistry, and U–Pb geochronology of Proterozoic mafic intrusions in the High Arctic: Relevance to the Nares Strait problem. Ph.D. thesis, University of Toronto, Toronto, Ont.

Denyszyn, S.W., Halls, H.C., and Davis, D.W. 2006. Paleomagnetism, geochemistry and U–Pb geochronology of orthogonal dykes in the Canadian High Arctic and Greenland, and tectonic implications. *In* Proceedings of the Geological Association of Canada – Mineralogical Association of Canada GAC–MAC) Joint Annual Meeting, May 2006, Montréal, Que., Canada. Program with Abstracts, Vol. 31, p. 37.

Denyszyn, S.W., Halls, H.C., and Davis, D.W. 2009. Paleomagnetism and U–Pb geochronology of Franklin dykes in High Arctic Canada and Greenland: a revised age and paleomagnetic pole, and evidence for rapid apparent polar wander in the Neoproterozoic. Canadian Journal of Earth Sciences, **46**: In press.

Ernst, R.E., and Buchan, K.L. 1998. Arcuate dyke swarms associated with mantle plumes on Earth: Implications for Venusian coronae. *In* Proceedings of the 29th Annual Lunar and Planetary Science conference, Houston, Tex. Abstract 1021.

Ernst, R.E., and Buchan, K.L. 2001. The use of mafic dike swarms

in identifying and locating mantle plumes, *In* Mantle plumes: their identification through time. *Edited* by R. E. Ernst and K. L. Buchan. Geological Society of America, Special Paper 352, pp. 247–265.

Ernst, R.E., Buchan, K.L., and Palmer, H.C. 1995. Giant dyke swarms: Characteristics, distribution and geotectonic applications. *In* Physics and chemistry of dykes. *Edited by* G. Baer and A. Heimann. Balkema, Rotterdam, The Netherlands, pp. 3–21.

Ernst, R.E., Buchan, K.L., Harris, J.A., Milner, M.W., and La Prairie, L.F. 2004. ^{40}Ar–^{39}Ar dating of the Lasard River mafic dykes, and implications for the focus of the 0.72 Ga Franklin Large Igneous Province of northern Canada. *In* Proceedings of the American Geophysical Union, Spring Meeting, May 2004, Montréal, Que., Canada. Program with Abstracts.

Ernst, R.E., Desnoyers, D.W., Head, J.W., andGrosfils, E.B. 2003. Graben–fissure systems in Guinevere Planitia and Beta Regio (264°–312°E 24°–60°N), Venus, and implications for regional stratigraphy and mantle plumes. Icarus, **164**: 282–316.

Fahrig, W.F. 1987. The tectonic settings of continental mafic dyke swarms: failed arm and early passive margin. *In* Mafic dyke swarms. *Edited by* H. C. Halls and W. F. Fahrig. Geological Association of Canada, Special Paper 34, pp. 331–348.

Fahrig, W.F., Irving, E., and Jackson, G.D. 1971. Paleomagnetism of the Franklin diabases. Canadian Journal of Earth Sciences, **8**: 455–467.

Féraud, G., Giannéri, G., and Campredon, R. 1987. Dyke swarms as paleostress indicators in areas adjacent to continental collision zones: Examples from the European and Northwest Arabian Plates, *In* Mafic dyke swarms. *Edited by* H. C. Halls and W. F. Fahrig. Geological Association of Canada, Special Paper 34, pp. 273–278.

Fleischer, R.L. 1982. Alpha-recoil damage and solution effects in minerals: uranium isotopic disequilibrium and radon release. Geochimica et Cosmochimica Acta, **46**: 2191–2201. doi:10.1016/0016-7037(82)90194-6.

Frisch, T. 1984*a*. Geology, Devon Ice Cap, District of Franklin, Northwest Territories. Geological Survey of Canada, Map 1574A, scale 1 : 250 000.

Frisch, T. 1984*b*. Geology, Mackinson Inlet, District of Franklin, Northwest Territories. Geological Survey of Canada, Map 1573A, scale 1 : 250 000.

Frisch, T. 1984*c*. Geology, Prince of Wales Mountains, District of Franklin, Northwest Territories. Geological Survey of Canada, Map 1572A, scale 1 : 250 000.

Frisch, T. 1988. Reconnaissance geology of the Precambrian shield of Ellesmere, Devon and Coburg Islands, Canadian Arctic Archipelago. Geological Survey of Canada, Memoir 409.

Grist, A.M., and Zentilli, M. 2005. The thermal history of the Nares Strait, Kane Basin, and Smith Sound region in Canada and Greenland: Constraints from apatite fission-track and (U–Th–Sm)/He dating. Canadian Journal of Earth Sciences, **42**: 1547–1569. doi:10.1139/e05-058.

Halls, H.C. 1986. Paleomagnetism, structure, and longitudinal correlation of Middle Precambrian dykes from northwestern Ontario and Minnesota. Canadian Journal of Earth Sciences, **23**: 142–157.

Halls, H.C., Campal, N., Davis, D.W., and Bossi, J. 2001. Magnetic studies and U–Pb geochronology of the Uruguayan dyke swarm, Rio de la Plata craton, Uruguay: Paleomagnetic and economic implications. Journal of South American Earth Sciences, **14**: 349–361. doi:10.1016/S0895-9811(01)00031-1.

Heaman, L.M., and LeCheminant, A.N. 1993. Paragenesis and U–Pb systematics of baddeleyite (ZrO_2). Chemical Geology, **110**: 95–126. doi:10.1016/0009-2541(93)90249-I.

Heaman, L.M., LeCheminant, A.N., and Rainbird, R.H. 1992. Nature and timing of Franklin igneous events, Canada: Implications for a Late Proterozoic mantle plume and the break-up of Laurentia. Earth and Planetary Science Letters, **109**: 117–131. doi:10.1016/0012-821X(92)90078-A.

Jackson, G.D., Ianelli, T.R., Narbonne, G.M., and Wallace, P.J. 1978. Upper Proterozoic sedimentary and volcanic rocks of northwestern Baffin Island. Geological Survey of Canada, Paper 78-14.

Jaffey, A.H., Flynn, K.F., Glendenin, L.E., Bentley, W.C., and Essling, A.M. 1971. Precision measurements of half lives and specific activities of ^{235}U and ^{238}U. Physical Review C: Nuclear Physics, **4**: 1889–1906.

Kigoshi, K. 1971. Alpha-recoil thorium-234: Dissolution into water and the Uranium-234/Uranium-238 disequilibrium in nature. Science, **173**: 47–48. doi:10.1126/science.173.3991.47. PMID: 17747313.

Kirschvink, J.L. 1980. The least-squares line and plane and the analysis of paleomagnetic data. Journal of the Royal Astronomical Society, **62**: 699–718.

Krogh, T.E. 1973. A low contamination method for hydrothermal decomposition of zircon and extraction of U and Pb for isotopic age determinations. Geochimica et Cosmochimica Acta, **37**: 485–494. doi:10.1016/0016-7037(73)90213-5.

Krogh, T.E. 1982. Improved accuracy of U–Pb zircon ages by the creation of more concordant systems using an air abrasion technique. Geochimica et Cosmochimica Acta, **46**: 637–649. doi:10.1016/0016-7037(82)90165-X.

Krogh, T.E., Corfu, F., Davis, D.W., Dunning, G.R., Kamo, S.L., Greenough, J., and Nakamura, E. 1987. Precise U–Pb isotopic ages of diabase dykes and gabbros using trace baddeleyite. *In* Mafic dyke swarms. *Edited by* H.C. Halls and W.H. Fahrig. Geological Association of Canada, Special Paper 34, pp. 147–152.

Lister, J.R., and Kerr, R.C. 1990. Fluid-mechanical models of dyke propagation and magma transport. *In* Mafic dykes and emplacement mechanisms. *Edited by* A. J. Parker et al. Balkema, Rotterdam, The Netherlands, pp. 101–104.

Ludwig, K.R. 2003. User's manual for Isoplot 3.00, a geochronological toolkit for Excel. Berkeley Geochronological Center, Berkeley, Calif., Special Publication 4.

Mattinson, J.M. 2005. Zircon U–Pb chemical abrasion ("CA–TIMS") method: Combined annealing and and multi-step partial dissolution analysis for improved precision and accuracy of zircon ages. Chemical Geology, **220**: 47–66. doi:10.1016/j.chemgeo.2005.03.011.

McElhinny, M.W., and McFadden, P.L. 2000. Paleomagnetism: Continents and oceans. 2nd ed. Academic Press, London, UK.

McKenzie, D., McKenzie, J.M., and Saunders, R.S. 1992. Dike emplacement on Venus and Earth. Journal of Geophysical Research, **97**(E10): 15 977 – 15 990.

Mège, D., and Ernst, R.E. 2001. Contractional effects of mantle plumes on Earth, Mars, and Venus. *In* Mantle plumes: their identification through time. *Edited* by R. E. Ernst and K. L. Buchan. Geological Society of America, Special Paper 352, pp. 103–140.

Meschede, M. 1986. A method of discriminating between different types of mid-ocean ridge basalts and continental tholeiites with the Nb–Zr–Y diagram. Chemical Geology, **56**: 207–218. doi:10.1016/0009-2541(86)90004-5.

Montési, L.G.J. 2001. Concentric dykes on the flanks of Pavonis Mons: Implications for the evolution of martian shield volcanoes and mantle plumes. *In* Mantle plumes: their identification through time. *Edited* by R. E. Ernst and K. L. Buchan. Geological Society of America, Special Paper 352, pp. 165–181.

Park, J.K. 1981. Paleomagnetism of basic intrusions from the

Brock Inlier, Northwest Territories, Canada. Canadian Journal of Earth Sciences, **18**: 1637–1641.

Pehrsson, S.J., and Buchan, K.L. 1999. Borden dykes of Baffin Island, Northwest Territories: A Franklin U–Pb baddeleyite age and a paleomagnetic reinterpretation. Canadian Journal of Earth Sciences, **36**: 65–73. doi:10.1139/cjes-36-1-65.

Romer, R.L. 2003. Alpha-recoil in U–Pb geochronology: Effective sample size matters. Contributions to Mineralogy and Petrology, **145**: 481–491. doi:10.1007/s00410-003-0463-0.

Schmidt, P.W. 1993. Paleomagnetic cleaning strategies. Physics of the Earth and Planetary Interiors, **76**: 169–178. doi:10.1016/0031-9201(93)90066-I.

Shellnutt, J.G., Dostal, J., and Keppie, J.D. 2004. Petrogenesis of the 723 Ma Coronation sills, Amundsen basin, Arctic Canada: Implications for the break-up of Rodinia. Precambrian Research, **129**: 309–324. doi:10.1016/j.precamres.2003.10.006.

Söderlund, U., and Johansson, L. 2002. A simple way to extract baddeleyite (ZrO_2). Geochemistry, Geophysics, Geosystems, **3**(2): 1014. doi:10.1029/2001GC000212.

Symons, D.T.A., and Sangster, D.F. 1992. Late Devonian paleomagnetic age for the Polaris Mississippi Valley type Zn–Pb deposit, Canadian Arctic Archipelago. Canadian Journal of Earth Sciences, **29**: 15–25.

Tokarski, A.K. 1990. Dyke swarms as stress indicators: Two constraints. *In* Mafic dykes and emplacement mechanisms. *Edited by* A. J. Parker et al. Balkema, Rotterdam, The Netherlands, pp. 101–104.

Torsvik, T.H., Smethurst, M.A., Meert, J.G., Van der Voo, R., McKerrow, W.S., Brasier, M.D., Sturt, B.A., and Walderhaug, H.J. 1996. Continental break-up and collision in the Neoproterozoic and Palaeozoic: A tale of Baltica and Laurentia. Earth-Science Reviews, **40**: 229–258. doi:10.1016/0012-8252(96)00008-6.

Trettin, H.P. 1991. Tectonic framework. *In* Geology of the Innuitian Orogen and Arctic Platform of Canada and Greenland. *Edited by* H. P. Trettin. Geological Survey of Canada, Ottawa, Ont., Geology of Canada, no. 3, pp. 59–66.

Walker, G.P.L. 1999. Volcanic rift zones and their intrusion swarms. Journal of Volcanology and Geothermal Research, **94**: 21–34. doi:10.1016/S0377-0273(99)00096-7.

Zijderveld, J.D.A. 1967. A. C. demagnetization of rocks: Analysis of results. *In* Methods in paleomagnetism. *Edited by* D. W. Collinson. Elsevier, Amsterdam, The Netherlands, pp. 254–286.

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